

The keyboard

There is no Ctrl key: Fn stands in. It is sticky, so tap **Fn**, let go, then the key.

Fn C	stop what is running
Fn Z	pause it (fg goes on)
Fn 1 4	screen 1 to 4
Fn ←	the screen before, after
Fn 6	brightness down, up
Fn 7 8	volume down, up
Fn 9	sound off and on
Fn 0	doze: dark, all kept (page 7)
Fn ↓	look back at what scrolled away
Fn ↑	start, end of the line
Fn Del	delete forwards
Fn L	clear the screen
Esc	clear the line; leave a program
Tab	finish a command or a name
↓	what you typed before
Sym	the symbols printed on the keys
Shift ×2	capitals until pressed again

On a USB keyboard, Ctrl is Ctrl, and Ctrl-A then 1-4 changes screen.

Screens and the shell

Four screens, each with its own shell, all running at once: music on 1, a game on 2, notes on 3.

TAB DOES THE TYPING

theme **cu** **Tab** theme cursor

play "0a" **Tab** the whole name, quoted

video **Tab** only clips, and folders

When there are several it lists them; type more and Tab again.

LOOKING BACK

Fn **A** then **↑** scrolls back through what went past; **↑** **↓** go on, **Esc** comes back. The bar shows 122: that many lines back.

SEVERAL AT ONCE

play **x.mp3** & runs behind; the shell is free

jobs what runs behind, or is paused

fg / **bg** to the front, or on behind

kill **%1** stop job 1

PLUMBING

```
ls -l | sort -k5 -n | tail -3
grep -rn exam ~/notes > found.txt
find ~ -name '*.mp3' | wc -l
```

Power and battery

THREE WAYS TO REST

Fn **0** **doze**: dark and quiet, all kept; a key and you are back where you were

suspend **sleep**: any key wakes it, as a fresh start; alarms still wake it

poweroff deepest: only **Enter** wakes it

Left alone for 15 minutes with nothing running, it goes to sleep by itself (power suspend 30m to change it).

CHARGING

Plug the charger module's own USB-C in: 1 A, full in about three hours. The board's port runs the machine from the PC but does not charge.

HOW MUCH IS LEFT

DOING	LASTS, ABOUT
reading, screen at half	1 day
music, screen dark	30 hours
video or games	12 hours
dozing	3 days
asleep	4 days and more

battery -v tells all, battery -l the log of starts and sleeps.

First steps

WAKING IT

Press any key. After three minutes without one the screen dims, after four it goes dark: the next key only lights it again.

A COMMAND

Type one and press **Enter**. Try these:

- ls** what is here
- cd music** go into a folder; cd alone goes home
- play song.mp3** play it
- cal** this month
- battery** how much is left

HELP

every command there is

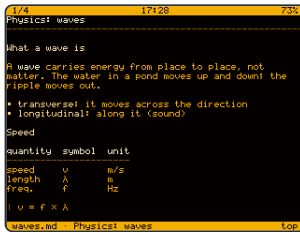
help play what one does, and how

man intro the manual, from the start

Tab finishes what you are typing

Notes

Put Markdown or text files in ~/notes (a folder for each subject) and notes lists them; pick one to read it, laid out for the screen: headings, lists, tables, quotes.



Space **B** next page, back

0 the headings, to jump to one

/ **N** find a word, and the next

+ **-** brighter, darker, for reading in bed

E open it in the editor

Q close it; it reopens where you left it

WRITING

edit FILE: **Fn** **S** save, **Fn** **Q** quit, **Fn** **F** find, **Fn** **0** go to a line, **Fn** **K** cut, **Fn** **U** paste, **Esc** all of it in a menu.

PART 1 OF 2 · THE MACHINE

PocketType

the pocket guide



Part 1 getting started, keys, the shell, power and battery, notes

Part 2 calendar, music, video, games, the internet, looks, every command

1.0 beta 4

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here.

How to read it: commands look like this: type them and press **Enter**. **Fn** **X** means tap Fn, let go, then tap X. help and man on the machine say more about everything.

pad on the Bluetooth radio
pad scan look for one (press its pairing button)
pad connect 1 join it; remembered
More games, in pico: snake, life (pico booklet).

A GAMEPAD
nes - q plays without sound.
q **Esc** quit
p a screenshot
space select
enter start
x **z** the A and B buttons
←→ **wasd** move
Nova the Squirrel, a free game by Novasquirrel.



Games
A NES inside: put .nes games in ~/roms and nes lists them.

Internet and files

WI-FI

wifi connect pick one from a list; remembered for next time

wifi scan networks in range, numbered

wifi what it is doing; **wifi off** saves power

Joining sets the clock too; so does **ntp**.

wget URL fetch a file

curl URL what a web address answers

ping HOST is it there?

passwd then telnet in from a PC, port 23

MOVING FILES

Quickest: the SD card into a PC. Or, with the USB cable, from the PC's pico-os folder:

```
make push FILE=notes.md
make pull FILE=/home/user/photos/shot-1.bmp
```

It goes beside the console: nothing is typed on the machine, and you can go on using it meanwhile.

Looks, and when things go wrong



theme list eleven themes, most dark only
theme gruvbox switch; dark or light after it
theme auto 7 20 light by day, dark at night
theme set bg #102030 a colour of your own
rotate 180 the screen the other way up

WHEN SOMETHING GOES WRONG

dark screen press a key; **Enter** if it was off

stuck program **Fn** **C**, or **Esc**

ps / kill N see and stop what runs

free, df memory, and room on the card

dmesg what the system has been saying

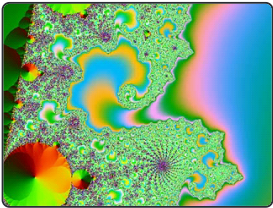
reboot start again (your files are safe)

factory-reset -y empties everything but Wi-Fi.

then copy the .pvt from clips/ into video/ on the SD card.

any video:
The machine plays its own kind of file, made on the PC from

MAKING CLIPS
q **Esc** stop, next time it goes on from there
s keep the picture, in ~/photos
0-9 0% to 90% of the way in
↑ a minute back, on
↓ 10 seconds back, on
← 10 seconds back, on
→ a minute back, on
space pauses: a bar shows where it is
video pick a clip from ~/video



Video and photos

Copy files into music/ on the SD card from a PC, or send one with make push FILE=song.mp3 DEST=~/.music/song.mp3 (page 14).

GETTING MUSIC ON
rec memo.wav until **Fn** **C**
rec -t 30 memo.wav thirty seconds
play memo.wav hear it
beep 880 200 a note, for scripts

RECORDING
play song.mp3 MP3, FLAC or WAV
play ~/.music/*.mp3 & a whole folder, behind
jobs, kill %1 what plays; stop it
volume 60 or **Fn** **7** **8**; **Fn** **9** mutes
Music plays on under whatever else you do, a game or a clip on another screen included: the sounds are mixed. Let the screen go dark, or doze, and it plays on for most of a day.

Music

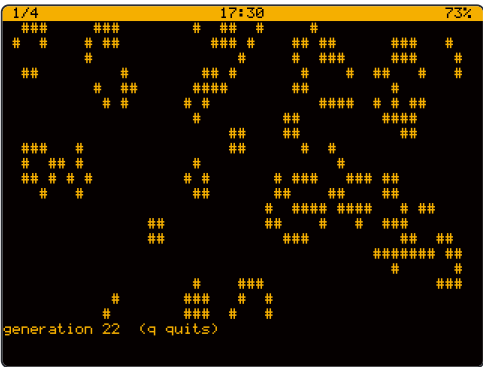
Every command

alarm alarms and timers
backlight brightness
battery what is left
beep a note
cal a month, a year
calendar the diary
cat more show a file
cd ls pwd folders
cp mv rm files
mkdir a new folder
curl wget the web
date date and time
df free room left
diff what changed
dmesg system log
edit the editor
find look for files
grep look inside them
head tail ends of a file
help man the manual
jobs fg bg jobs
kill ps top processes
led the charge light
nes pad games
notes read notes
ntp set the clock
pico picoc the language
ping is it there
play rec sound
power sleep times
poweroff switch off
reboot start again
rotate turn the screen
screenshot keep the screen
sed sort change, order text
suspend sleep now
theme colours
todo the to-do list
uptime how long it's up
video view clips, pictures
volume loudness
wc count words
wifi the network
help lists all of them, and **help NAME** says how to use one.

PART 2 OF 2 · APPS AND MORE

PocketType

the pocket guide



Part 2 calendar and to-do, alarms, music, video and photos, games, Wi-Fi and files, themes, when something goes wrong, and every command on the back

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ALARMS
alarm 7:00 mo-fr up every weekday
alarm 25m tea a timer
alarm what is set; **alarm rm 2**

todo the list, to tick off
todo add milk @fri or **@today, @30/9**
todo done 1 tick one off

TO DO
calendar agenda
calendar add 30/9 10:00 dentist i30 chimes half an hour before; every mo, we and yearly
2/10 repeat

calendar
the month:
arrows move, **A**
adds, **D** deletes
the week ahead



Diary

10
Rings even from sleep.
Kinging, any key snoozes it nine minutes and **Esc** stops it. It